Answer Key

Exercise - 1.1



1. Right pair

Bharti Krishnateertha – Vedic Mathematics
Varahamihira – Panch Siddhanta
Brahmagupta – Brahmasphoot Siddhanta
Bhaskaracharya – Siddhanta Shiromani
Aaryabhatt – Aaryabhatiya

2. (1) b (2) 10^{53}
3. Aaryabhatt, Aaryabhatiya
4. Beej Ganit
5. 16

Exercise - 1.2



2. 6, 2, 7, 9, 7, 3, 1, 2, 4, 4,
3. 2, 3, 5, 7

Exercise - 1.3



1. 736
2. 2288
3. 3404
4. 3025
5. 39483
6. 96048
7. 311472
8. 367836

Exercise - 1.4



1. 5643
2. 43775622
3. 86913
4. 34499655
5. 432
6. 9447543

Exercise - 1.5



1. 221
2. 616
3. 1224
4. 9009
5. 1225
6. 2016
7. 5616
8. 9021
9. 11024
10. 11025

11. 42021 12. 164025 13. 255016 14. 366021 15. 497024
16. 819025 17. 38021 18. 87016 19. 156016 20. 245025

Exercise - 1.6



1. 169 2. 10608 3. 11130 4. 9212
5. 12444 6. 10272 7. 1016048 8. 972052 9. 1013968.

Exercise - 1.7



225, 625, 1225, 2025, 3025, 5625, 7225, 9025, 11025, 13225.

Exercise - 1.8



(1) 1156 (2) 361 (3) 2916 (4) 4096 (5) 8464.

Exercise - 1.9



114, 196, 10404, 11025, 11664, 8836, 992016

Exercise - 1.10



(1) 97 (2) 87 (3) 91 (4) 57

Exercise - 1.11



(1) $4x^2 + 21x + 5$ (2) $12x^2 + 18xy + 6y^2$ (3) $x^2 - 9y^2$
(4) $x^2 + 4x + 16$ (5) $x^4 + 5x^3 + 11x^2 + 11x + 4$
(6) $6x^4 + 17x^2y - 2x^2 + 12y^2 - y - 20$

Exercise - 2.1

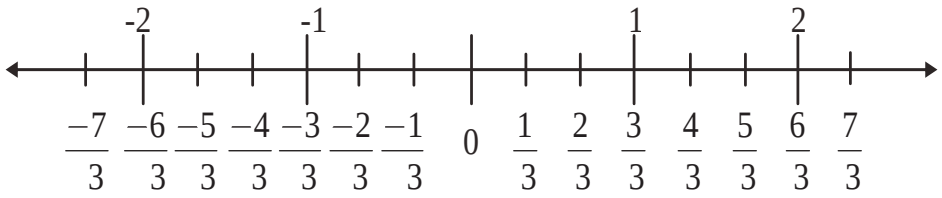


1. (i) 1 (ii) Zero (iii) Zero (**Note :-** There can be several examples of (i) and (iii))

2. $\frac{25}{4}, \frac{26}{4}, \frac{27}{4}$ 3. $\frac{25}{42}, \frac{26}{42}, \frac{27}{42}, \frac{28}{42}, \frac{29}{42}$

4. $\frac{33}{48}, \frac{34}{48}, \frac{35}{48}$ 5. Any four numbers between $\frac{-4}{5}, \frac{-3}{5}, \frac{-2}{5} \ldots\ldots \frac{8}{5}, \frac{9}{5}$

6. Any three numbers between $\frac{21}{40} \ldots\ldots\ldots\ldots \frac{35}{40}$

7. 

8. (i) 25.2 (Terminating decimal)
(ii) 20.9375 (Terminating decimal)
(iii) 3.14285714... (Non terminating recurring)
(iv) –39.33... (Non terminating recurring)

9. (i) $\frac{53}{100}$ (ii) $\frac{84}{5}$ (iii) $\frac{421}{4}$ (iv) $\frac{184}{25}$

10. (i) $\frac{70}{99}$ (ii) $\frac{277}{33}$ (iii) $\frac{563}{180}$ (iv) $\frac{5120}{999}$

Exercise - 2.2



1. (i) $18\sqrt{3}$ (ii) $3\sqrt{7}$

2. $7\sqrt{2}+\sqrt{5}$ 3. $3\sqrt{7}-8\sqrt{5}$

4. (i) 1 (ii) 20 (iii) $7+3\sqrt{6}$ (iv) $10-2\sqrt{21}$

5. (i) $\frac{\sqrt{5}}{5}$ (ii) $\frac{\sqrt{6}}{3}$ (iii) $\frac{\sqrt{7}}{1}$

6. (i) $a=\frac{13}{11};\ b=\frac{4}{11}$ (ii) $a=4,\ b=1$

7. (i) $20+3\sqrt{3}$ (ii) $16-\sqrt{3}$ 8. 6

Exercise - 3.1



1. (i) 32 (ii) 81 (iii) 16

2. (i) 8 (ii) 625 (iii) –9765625

3. (i) $128t^3$ (ii) -1

5. (i) 8.52×10^{-12} (ii) 8.02×10^{15} (iii) 4.196×10^{10}

6. (i) 0.00000502 (ii) 0.00000007 (iii) 1000010000

7. (i) 7×10^{-6} meter (ii) 1.2756×10^{7} (iii) 8×10^{-2} meter

Exercise - 3.2



1. (i) 4 (ii) 3 (iii) 5

2. (i) 529 (ii) 1,331 (iii) 441 (iv) $\frac{1}{15}$

3. (i) $\left(\frac{25}{9}\right)^{\frac{4}{3}}$ (ii) $\left(\frac{13}{2}\right)^{\frac{1}{3}}$ (iii) $\frac{1}{3}$ (iv) 243

4. (i) $\sqrt[9]{512} < \sqrt[3]{64} < \sqrt{81}$ (ii) $\sqrt[4]{625} < \sqrt[3]{343} < \sqrt{100}$

(iii) $\sqrt[5]{243} < \sqrt[3]{216} < \sqrt{64}$ (iv) $\sqrt[7]{128} < \sqrt[4]{256} < \sqrt[3]{1000}$

5. (i) Surd (ii) No (iii) Surd

(iv) No (v) Surd

Exercise - 4.1



1. (i) It is polynomial because the power of variable is whole number.

(ii) It is not a polynomial, because we writes $z + \frac{3}{z}$ as $z + 3z^{-1}$

i.e. the power of z is not a whole number.

(iii) It is not a polynomial, because we writes $\sqrt{y} + 2y + 3$ as $y^{\frac{1}{2}} + 2y + 3$

i.e. the power of y is $\frac{1}{2}$ which is not the whole number.

(iv) It is polynomial, because the power of variable is a whole number.

(v) It is polynomial, because the power of variable is a whole number.

2. (i) 5 (ii) 3 (iii) -5 (iv) $\frac{1}{2}$ (v) $\frac{1}{4}$

3. (i) $\frac{1}{5}$, 5 (ii) $\sqrt{2}$, 7 (iii) 0, 2

4. (i) $x^4 + 3$, $3x^4 + 4$ etc. (ii) $y^6 + 3y + 3$, $t^6 + 3t - 5$ etc.
(iii) $3x^5$, z^5, $11z^5$ etc.

5. (i) 3 (ii) 9 (iii) 4
(iv) 3 (v) 1 (vi) 0

6. Constant (vi), (x), (xii) Linear (iv), (ix), (xi)
Binomial (iii), (vii), (viii) Trinomial (i), (ii), (v)

Exercise - 4.2



1. (i) -2 (ii) 4 (iii) -56

2. (i) 2, 3, 36, 119 (ii) -1, 0, 3, 8
(iii) $\frac{1}{3}, \frac{4}{3}, \frac{7}{3}, \frac{16}{3}$ (iv) 1, 0, 9, 28 (v) 2, 1, 4, 5

3. (i) Yes (ii) Yes (iii) No (iv) Yes
(v) No (vi) Yes (vii) Yes

4. (i) -6 (ii) 6 (iii) 0 (iv) 0 (v) $\frac{-d}{c}$
(vi) 2 (vii) $-\frac{3}{2}$ (viii) $\sqrt{5}$ (ix) $\frac{4}{3}$

Exercise - 4.3



1. (i) $5x^2 + 5x + 6$ (ii) $3p^3 + 8p^2 - 7p + 11$ (iii) $2x^3 + 7x^2 - 8x - 1$

2. (i) $4y^2 + 9y$ power 2 (ii) $-r^2 + 4r + 17$ power 2
(iii) $3x^3 + 4x^2 + 8$ power 3

3. (i) $7t^3 - 4t^2 + 5t$ (ii) $-4p^3 + 7p^2 + 8p - 12$ (iii) $7z^3 - 10z^2 + 9z + 17$

4. $2x^4 + 2x^3 - 3x^2 + 11x + 4$ 5. 5

6. $-x^3 + 4x^2 - x - 2$ 7. $-u^7 + 4u^6 - 4u^2 - u - 6$

8. $-3y^2 - y + 1$ 9. $-t^3 - 2t - 11$

10. (i) $21x^3 + 34x^2 + 11x + 4$ (ii) $15x^5 - 45x^4 + 36x^3 - 18x^2 + 12x$
(iii) p^7 & $5p^5 + p^4 + 3p^3$ &$5p^2 + 3$

11. $2x^6 + 14x^4 + 3x^3 - 21x - 9$ 12. 5

Exercise - 5.1



1. (i) $x + (x + 1) = 11$ (ii) $2y + y = 30$ (iii) $2z + 2z + z = 40$
   (iv) $2[(w + 3) + w] = 15$ (v) $2x + 3x + 4x = 18$

2. (i) $x = 3$ (ii) $p = \frac{23}{5}$ (iii) $x = 2$
   (iv) $t = -\frac{1}{9}$ (v) $z = 1$ (vi) $x = \frac{2}{7}$

3. (i) If we add 3 in a certain number, we get 27.
   (ii) Sneha's age is half at Rajiv's age and sum of both their ages is 18 years.
   (iii) If we add 2 in any number, then divide the sum by same number, we get 30.

4. (i) $\frac{-6}{5}$ (ii) $k = 4$ (iii) $p = 5$
   (iv) $x = -\frac{68}{25}$ (v) $m = \frac{7}{5}$ (vi) $t = 2$
   (vii) $x = \frac{27}{10}$ (viii) $x = \frac{35}{33}$ (ix) $y = -8$

Exercise - 5.2



1. 6 meter 2. 15 *cm*, 10 *cm* 3. $35^o, 50^o, 95^o$
4. 3, 5, 7 5. 20 *cm* 6. 35
7. 28 and 20 years 8. 15 9. 80, 120
10. 180, 185, 190 11. Rohit = 39 years, Pradeep = 17 years
12. 22 *km / hour*

Exercise - 6.1



1. (i) A = 6, B = 9 (ii) X = 5, Y = 6
   (iii) L = 0, M=1, N = 8 (iv) Z = 5
   (v) X = 1, Y = 4 (vi) P = 4, Q = 7
   (vii) M = 7, L = 4

Exercise - 6.2



1. Multiple of 5 = 9560, 205, 800, Multiple of 10 = 9560, 800
2. 1, 4, 7
3. P = 0 or 5
4. A = 7, B = 2; A = 3, B = 6; A = 2, B = 7; A = 6, B = 3
5. (i) Yes (ii) No (iii) Yes (iv) Yes
6. (i) Yes (ii) No (iii) Yes (iv) Yes
7. (i) Yes (ii) Yes (iii) Yes (iv) No

Exercise - 7.1



1. Rs. 15800
2. 2 : 3
3. Rs. 190 and 20%
4. Rs. 8050
5. Rs. 139100
6. Rs. 1900
7. Equal
8. (i) Rs. 210 (ii) Rs. 157.50
9. Rs. 5000
10. Rs. 9500

Exercise - 7.2



1. (i) Compound interest = Rs. 3248.70, Principal = Rs. 10248.70
   (ii) Compound interest = Rs.1590, Principal = Rs. 7840
   (iii) Compound interest = Rs. 2522, Principal = Rs. 18522
2. Amount = Rs. 175616, interest amount = Rs. 50616
3. Rs. 16125
4. Rs. 3.15
5. Rs. 80
6. Rs. 374.59
7. Rs. 741.90
8. Rs. 544.69
9. Rs. 30250
10. Rs. 609
11. Rs. 6000
12. Rs. 16000
13. 4%
14. 5%
15. 3 years
16. $1\frac{1}{2}$ years
17. Rs. 16000

Exercise - 7.3



1. Rs. 33.80
2. 33.8%
3. Rs. 673.11
4. Rs. 8302.075
5. Rs. 639.35
6. Rs. 6250
7. Rs. 4840
8. Rs. 1313.30
9. Rs. 12696

Exercise - 8.1



1. (i) $\sin A = \frac{4}{5}, \cos C = \frac{4}{5}, \tan A = \frac{4}{3}$

(ii) $\sin A = \frac{5}{13}, \cos C = \frac{5}{13}, \tan A = \frac{5}{12}$

(iii) $\sin A = \frac{12}{13}, \cos C = \frac{12}{13}, \tan A = \frac{12}{5}$

(iv) $\sin A = \frac{12}{15}, \cos C = \frac{12}{15}, \tan A = \frac{12}{9}$

Exercise - 8.2



1. (i) $\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}, \cot\theta = \frac{4}{3}, \sec\theta = \frac{5}{4}, \operatorname{cosec}\theta = \frac{5}{3}$

(ii) $\cos\theta = \frac{12}{13}, \tan\theta = \frac{5}{12}, \cot\theta = \frac{12}{5}, \sec\theta = \frac{13}{12}, \operatorname{cosec}\theta = \frac{13}{5}$

(iii) $\sin\alpha = \frac{2\sqrt{2}}{3}, \tan\alpha = \frac{2\sqrt{2}}{1}, \cot\alpha = \frac{1}{2\sqrt{2}}, \sec\alpha = 3, \operatorname{cosec}\alpha = \frac{3}{2\sqrt{2}}$

(iv) $\sin\theta = \frac{1}{\sqrt{2}}, \cos\theta = \frac{1}{\sqrt{2}}, \tan\theta = 1, \sec\theta = \sqrt{2}, \operatorname{cosec}\theta = \sqrt{2}$

(v) $\sin A = \frac{4}{5}, \cos A = \frac{3}{5}, \tan A = \frac{4}{3}, \cot A = \frac{3}{4}, \sec A = \frac{5}{3}$

(vi) $\sin\beta = \frac{\sqrt{3}}{2}, \cos\beta = \frac{1}{2}, \tan\beta = \sqrt{3}, \cot\beta = \frac{1}{\sqrt{3}}, \operatorname{cosec}\beta = \frac{2}{\sqrt{3}}$

(vii) $\sin A = \frac{1}{\sqrt{10}}, \cos A = \frac{3}{\sqrt{10}}, \tan A = \frac{1}{3}, \cot A = 3, \sec A = \frac{\sqrt{10}}{3}$

2. $\frac{420}{841}$ 3. $\frac{22}{45}$ 4. $\frac{8}{31}$

5. $\frac{2\sqrt{2}}{3}, 3 + \frac{1}{2\sqrt{2}}, \frac{3}{2\sqrt{2}}$ 6. 1 7. 7 8. $\frac{1}{2}$

Exercise - 8.3



1. (i) c (ii) c

2. (i) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (ii) $\sqrt{\frac{2}{3}}+2$ या $2+\frac{\sqrt{6}}{3}$ या $\frac{6+\sqrt{6}}{3}$ (iii) 3

   (iv) $\frac{1}{2-\sqrt{3}}$ (v) 4 (vi) $\frac{\sqrt{3}}{2}$ (vii) $\frac{1}{3}$ (viii) $\frac{3}{2}$

3. (i) False (ii) False (iii) True (iv) True (v) True

Exercise - 8.4



1. 45° 2. 60° 3. 30° 4. 30°
5. 60° 6. 0° 7. 60° 8. 60°
9. 60° 10. 0° 11. 30°

Exercise - 9.1



1. $x = 130^\circ$, $y = 50^\circ$
2. $\angle QOT = 30^\circ$ and reflex $\angle ROT = 250^\circ$
3. $z = 126^\circ$
5. If $a + b = c + d$ will be on a line then $\frac{360^\circ}{2} = 180^\circ$
7. $\angle ABY = 122^\circ$, reflex $\angle YBX = 302^\circ$

Exercise - 9.2



1. (i) $x = 36^\circ$, $y = 108^\circ$
   (ii) $x = 29^\circ$, $y = 87^\circ$
   (iii) $x = 95^\circ$, $y = 35^\circ$
2. $x = 126^\circ$ 3. $\angle AGE = 126^\circ$, $\angle GEF = 36^\circ$, $\angle FGE = 54^\circ$
4. 60° 5. $x = 50^\circ$, $y = 77^\circ$ 6. $x = 80^\circ$, $y = 100^\circ$
7. (i) $x = 59^\circ$, $y = 60^\circ$ (ii) $x = 40^\circ$, $y = 40^\circ$
   (iii) $x = 18^\circ$, $y = 60^\circ$ (iv) $x = 20^\circ$, $y = 63^\circ$

8. (i) 140° (ii) 100° (iii) 250°

10. Strike on mirror and then make the alternate angle $\angle ABC = \angle BCD$, hence $AB \parallel CD$.

Exercise - 9.3



1. $\angle ACB = 17^o$, $\angle ABC = 129^o$, $\angle DBC = 51^o$

2. (i) $x = 40^o$, $y = 70^o$ (ii) $x = 50^o$, $y = 20^o$
   (iii) $x = 51^o$, $y = 35^o$ (iv) $x = 30^o$, $y = 75^o$

3. $\angle 1 = 60^o$ 4. $\angle PRQ = 65^o$ 5. $\angle OZY = 32^o$, $\angle YOZ = 121^o$

6. $\angle DCE = 92^o$ 7. $\angle SQT = 60^o$ 8. $x = 53^o$, $y = 37^o$

Exercise - 10.1



1. b 2. b 3. c 4. b 5. c

6. (i) R.H.S. (ii) SAS (iii) ASA
   (iv) AAS (v) SSS

7. 25 8. 160 meter (SSS congruency rule)

9. Yes (This is a rectangular shape and the opposite side are equal in rectangle).

10. From ΔABC and ΔACD, BC = AD and AB = CD
    Then AC = CA (common side)
    $\Delta ABC \cong \Delta ACD$

Exercise - 10.2



1. (iii) 2. (i) 3. (iv) 4. (iii) 5. (iii)

6. 30° 7. 23°, acute angled triangle

8. From ΔABC and ΔDEF
   BC = EF, $\angle B = \angle E$, $\angle C = \angle F$
   $\Delta ABC \cong \Delta DEF$

9. From ΔABD and ΔABC
   AD = BC, $\angle A = \angle B$, $AB = BA$
   $\Delta ABD \cong \Delta ABC$
   $\therefore BD = AC$, $\angle ABD = \angle BAC$

Exercise - 10.3



1. $\angle B < \angle A \quad \therefore AO < BO \quad \ldots\ldots (1)$

   $\angle C < \angle D \quad \therefore OD < OC \quad \ldots\ldots (2)$

   equation (1) + (2)

   $\therefore AO + OD < BO + OC$

   $AD < BC$

2. $\angle x > \angle y$

   $\therefore \angle PMN = 180^\circ - \angle x, \quad \angle PNM = 180^\circ - \angle y$

   $\therefore \angle PNM > \angle PMN$ (The larger side opposite to larger angle)

   $\therefore PM > NP$

3. $PQ > PR$

   $\angle PRQ > \angle PQR$

   $\angle QRS > \angle RQS$

   $SQ > SR$

4. $\angle PQR = \angle PRQ$

   $\therefore \angle PQS > \angle PRQ$

   $\therefore PS > PQ$

Exercise - 11.1



2. Angles made at point C and R on a line AB and CD are half of 180° i.e. 90°. These are opposite angles therefore the figure is rectangle.

3. (i) Alternate angle (ii) ABCD is a parallelogram

4. $\angle P = \angle Q = 90^\circ$

   $DP = QB$

   $AB = DC$

   $\therefore \Delta APB \cong \Delta CQD$

Exercise - 14.1



1. 110 *cm* 2. 55.44 *sq cm* 3. 616 *sq* meter

4. 1100 meter 5. 9 meter 6. 168 meter

7 Rs. 44,785.71 8. 25.6 *cm*, 269.5 *sq cm*

9. $\theta = 60°$ 10. $42\sqrt{2}$ *cm*

Exercise - 15.1



1. 3375 cubic meter 2. 8 *cm*

3. (i) 4 times (ii) 9 times (iii) $n \times n$ times or n^2 times

4. 6 meter

5. (i) cubical peti, 40 *sq cm* (ii) cubical peti, 10 *sq cm*

6. 3 days 7. 4000 cubic *cm* 8. 12288 cubic *cm*

9. 350 *sq cm* 10. 6 meter 11. 4167 bricks

12. Rs. 4320 13. 16000 14. 6*cm*, 4:1 15. 7 sheet

Exercise - 16.1



1. Class interval – width of class
Size of a class – the difference of class mark of two consecutive classes
Class mark – digit of class
Class frequency – frequency
Class limits – higher and lower limit of a class.

2. Inclusive class – the upper limit of a class should not be equal to lower limit of a next class.
Exclusive class – the upper limit of a class should be lower limit of next class.

3.

Maximum Temp. (Month of August)	Tally Marks	Frequency
28.5 – 29.5	|	1
29.5 – 30.5	|	1
30.5 – 31.5	| |	2
31.5 – 32.5	| |	2
32.5 – 33.5	𝍸 | |	7
33.5 – 34.5	| | | |	4
34.5 – 35.5	| | | |	4
35.5 – 36.5	| |	2
36.5 – 37.5	𝍸 | |	7
37.5 – 38.5	|	1

4.

Distance (in *km*)	Tally Marks	Frequency
0 — 5	𝍸	5
5 — 10	𝍸 𝍸 |	11
10 — 15	𝍸 𝍸 ||	12
15 — 20	𝍸 ||||	9
20 — 25	|	1
25 — 30	|	1
30 — 35	|	1

5.(i)

Marks	Tally Marks	Frequency
0	||	2
1	𝍸	5
2	𝍸	5
3	𝍸 |||	8
4	||||	4
5	𝍸	5
6	||||	4
7	||||	4
8	𝍸	5
9	𝍸 |||	8

(ii) Zero (iii) 3 and 9

6.

Production (in quintals)	10—15	15—20	20—25	25—30	30—35	35—40	40—45	45—50
Frequency	1	7	10	8	7	2	2	3

7.(i)

Hours of TV programme	Tally Marks	Frequency
0 — 5	𝍸 |||	8
5 — 10	𝍸 𝍸	10
10 — 15	𝍸 |	6
15 — 20	|||	3
20 — 25	|||	3
25 — 30	||||	4
30 — 35	||||	4
35 — 40	||	2

(ii) 0 (iii) 15-20 (iv) 32.5 (v) 13

Exercise - 16.2



1. (i) b (ii) c (iii) c
   (iv) b (v) a

2.(i)

Time (in second)	Frequency
15 – 25	3
25 – 35	5
35 – 45	5
45 – 55	8
55 – 65	4

(ii) —— Histogram
(iii) ..... Frequency Polygon

